Foundations: Population Methods

Swarm theory

15 July 2026

Contents

1 Part 1 — Deep Foundation & Core Swarm Pillars

Self-contained theory of population-based metaheuristics, written to be defended line-by-line in a viva and to seed the arXiv "Learning Journey" report. Notation is fixed once in §1.1 and reused throughout. Every update equation is attributed to its seminal source.

1.1 1.1 What a population-based metaheuristic *is* (formal template)

1.1.1 1.1.1 The optimization problem

We are given a search space $\mathcal{S}$ and an objective (fitness) function

$$f : \mathcal{S} \to \mathbb{R}, \qquad \text{find } x^\star \in \arg\min_{x \in \mathcal{S}} f(x)$$

(maximization is the sign-flip $f \mapsto -f$; our Lab-3 code maximizes coverage, so we negate mentally where needed). $\mathcal{S}$ may be continuous ($\mathcal{S} \subseteq \mathbb{R}^n$, bounded by a box $[\ell, u]$), combinatorial (permutations, binary strings $\{0,1\}^n$), or mixed. We assume only that f can be **evaluated pointwise** — an *oracle* or *black box*. We do **not** assume f is convex, differentiable, continuous, or even given in closed form.

A **metaheuristic** is a problem-independent, stochastic, iterative strategy that samples $\mathcal{S}$, guided only by the fitness values it has observed, to drive the best-so-far sample toward $x^\star$. A **population-based** metaheuristic maintains not one incumbent but a *set* (population) of μ candidate solutions simultaneously.

1.1.2 1.1.2 State, operators, and the generic template

Let the **population** at iteration t be

$$P_t = (x_t^{(1)}, x_t^{(2)}, \dots, x_t^{(\mu)}) \in \mathcal{S}^\mu.$$

Every algorithm in this dossier (GA, PSO, ACO, ABC, FA, GWO, CS, WOA) is an instance of the same three-operator loop:

1. **Evaluate** — compute $f(x_t^{(i)})$ for all i (the only place f is touched; the *budget* is counted here).
2. **Vary** — produce candidate offspring/moves by a stochastic operator $\mathcal{V}$ that reads the current population (and possibly external memory M_t such as pheromone or personal bests):

$$\tilde{P}_t \sim \mathcal{V}(\,\cdot \mid P_t, M_t).$$

   $\mathcal{V}$ is where the algorithms differ: recombination + mutation (GA), velocity update (PSO/FA), pheromone-biased construction (ACO), neighbour perturbation (ABC), leader-guided steps (GWO), Lévy flights (CS), spiral/encircling (WOA).
3. **Select** — choose the survivors P_{t+1} from $P_t \cup \tilde{P}_t$ by a rule $\mathcal{Q}$ biased toward lower f (fitter):

$$P_{t+1} = \mathcal{Q}(P_t, \tilde{P}_t).$$

```
initialise P_0 ~ Uniform(S),  M_0 = empty
evaluate f(P_0)
for t = 0, 1, 2, ... until stop:
    P~ ← Vary(P_t, M_t)         # stochastic reproduction / movement
    evaluate f(P~)              # <-- one "generation" of budget
    P_{t+1} ← Select(P_t, P~)    # fitness-biased survival
    M_{t+1} ← UpdateMemory(...)  # pheromone / pbest / gbest / leaders
return best-so-far x*
```

The engine is **variation ⊕ selection**: variation *injects entropy* (proposes unseen points), selection *removes entropy* (concentrates probability mass on good regions). Optimization is the controlled tension between the two — this is the exploration/exploitation axis formalized in §1.4.

1.1.3 1.1.3 The Markov-chain view (why this is analyzable)

Because P_{t+1} depends only on (P_t, M_t) and fresh randomness, the sequence of populations is a **Markov chain** on the (finite or measurable) state space $\Omega = \mathcal{S}^\mu \times \mathcal{M}$:

$$\Pr\left(P_{t+1} = q \mid P_t = p, P_{t-1}, \dots\right) = \Pr\left(P_{t+1} = q \mid P_t = p\right) =: K(p, q),$$

where K is the transition kernel induced by $\mathcal{V}$ and $\mathcal{Q}$. For a finite $\mathcal{S}$ (e.g. bitstrings), K is a stochastic matrix and the whole apparatus of Markov-chain theory (irreducibility, ergodicity, absorbing states) applies — this is exactly how convergence is *proved* (§1.6). Two structural facts follow immediately:

- If mutation assigns **strictly positive probability** to every point of $\mathcal{S}$ from every state (global reachability), the chain can never be permanently trapped: $\Pr(\exists\, t : x_t^{(i)} = x^\star) \to 1$. This is the *global-search property*.
- If selection is **elitist** (the best-so-far is never discarded), the best-so-far fitness is a monotone, bounded process and therefore converges (Rudolph 1994; §1.6.2).

1.1.4 1.1.4 Generational vs. steady-state

- **Generational**: the entire population is replaced each iteration (P_{t+1} built from $\tilde{P}_t$). Canonical GA, PSO, WOA, GWO, FA are generational.
- **Steady-state**: only one/few individuals are replaced per step (e.g. replace-worst). ABC is effectively steady-state per food source; ACO updates pheromone incrementally.
- **Elitism** is an orthogonal switch: retain the top-k verbatim. It converts a merely *exploratory* chain into a *convergent* one (§1.6.2) at the cost of selection pressure (§1.5). Our PSO keeps `gbest`, an implicit elitism of size 1.

1.2 1.2 Why and when they beat gradient-based and exact solvers

1.2.1 1.2.1 What gradient methods assume — and when it breaks

Gradient descent, Newton, quasi-Newton (L-BFGS), and conjugate-gradient all update

$$x_{k+1} = x_k - \eta_k \, H_k^{-1} \, \nabla f(x_k)$$

and therefore **require** (i) f differentiable with a computable (or finite-difference-approximable) ∇f, and (ii) local gradient information to be *informative* about the global optimum. They are *local* methods with (super)linear convergence **on convex or locally convex, smooth** landscapes. They fail — provably or practically — when:

- **No gradient exists**: f is discontinuous, piecewise-constant, or defined by a simulation/`max`/threshold. *Our Lab-3 objective* uses $\max_j \mathrm{RSSI}_{ij}$ (a kink), a logistic soft-threshold, and integer **wall-crossing counts** — the last makes f literally piecewise-constant in AP position, so $\nabla f = 0$ almost everywhere and undefined on the kinks. A gradient step has *nothing to descend*.
- **Non-convex / multimodal**: many local minima; gradient descent converges to whichever basin it starts in. Restarts help only $\propto$ (number of basins)$^{-1}$.
- **Ill-conditioned or deceptive**: the gradient points *away* from the global optimum (deceptive functions, §1.7.4).
- **Black-box, expensive, noisy**: finite-difference gradients cost $n+1$ evaluations per step and amplify noise.

1.2.2 1.2.2 What exact/deterministic solvers assume — and when it breaks

Exact methods (branch-and-bound, MILP, dynamic programming, exhaustive enumeration, A*/backtracking as in Labs 1–2) return a *provably* optimal solution but with worst-case cost that **explodes**:

- Combinatorial spaces grow factorially/exponentially: k-of-G placement has $\binom{G}{k}$ options; a permutation of n has $n!$. Our Grid-Search baseline is exactly this — a $\binom{25}{3} = 2300$-way enumeration of a *coarse* grid; a grid fine enough to rival a continuous optimizer needs $\binom{G}{k} \to \infty$. This is the **curse of dimensionality** made concrete.
- Continuous global optimization is NP-hard in general; exact solvers need special structure (linearity, convexity, submodularity) that black-box f does not provide.

1.2.3 1.2.3 The regime where metaheuristics win — and where they lose (honest)

Population metaheuristics are the rational choice when **all** of these hold: no reliable gradient; non-convex/multimodal or combinatorial; black-box or simulation-based f; a "good enough, fast enough" near-optimum is acceptable in place of a certificate of optimality; and the per-evaluation cost permits thousands of samples.

They **lose** — and one must say so in a viva — when:

- f is **convex and smooth**: gradient/Newton converge (super)linearly to the *global* optimum; a swarm is slower and gives no certificate.
- f is **low-dimensional and cheap**: dense grid or multi-start local search dominates.
- The problem has **exploitable structure** (LP/MILP/convex/DP/submodular): an exact solver returns a *provably* optimal answer with a bound; a metaheuristic cannot.
- Evaluations are **extremely expensive** ($\sim$minutes each): Bayesian optimization / surrogate models use the budget better than a 30-particle swarm.

Metaheuristics buy **robustness to landscape pathology** at the price of **optimality guarantees**. That trade is the thesis of this lab.

1.3 1.3 The No Free Lunch (NFL) theorem

Source. D. H. Wolpert & W. G. Macready, "No Free Lunch Theorems for Optimization," *IEEE Transactions on Evolutionary Computation* 1(1):67–82, 1997, DOI 10.1109/4235.585893.

Setup. Fix finite $\mathcal{S}$ and finite value set $\mathcal{Y}$. An algorithm a generates a time-ordered sample (trace) $d_m = ((x_1, y_1), \ldots, (x_m, y_m))$ of m distinct visited points; let d_m^y be the sequence of *fitness values* seen. Performance is any functional $\Phi(d_m^y)$ (e.g. best value found).

Statement (NFL-1). For any two algorithms a_1, a_2, averaged **uniformly over all possible objective functions** $f : \mathcal{S} \to \mathcal{Y}$,

$$\sum_f \Pr\left(d_m^y \mid f, m, a_1\right) \;=\; \sum_f \Pr\left(d_m^y \mid f, m, a_2\right).$$

Consequently $\sum_f \Phi(d_m^y \mid f, m, a_1) = \sum_f \Phi(d_m^y \mid f, m, a_2)$ for **every** performance measure Φ: *summed over all functions, all non-revisiting algorithms are identical.* Any above-average performance on one class of f is paid for by exactly equal below-average performance on the complementary class. (An analogous NFL holds for supervised learning.)

Why it is true, in one line. The uniform average over all f makes the histogram of unseen fitness values independent of *how* points were chosen; an algorithm's cleverness is about *which point to sample next*, but under the uniform prior the value at any unseen point is equally likely to be anything, so ordering gains nothing.

Implications for algorithm selection.

1. **There is no universally best optimizer.** Claims that "algorithm X dominates" are meaningful only *relative to a function class*. Benchmarking on a biased suite (e.g. only separable, smooth functions) and generalizing is an NFL fallacy.
2. **Performance is bought with matched priors.** An algorithm beats another *iff* its inductive bias matches the structure of the target functions. GWO's leader-hierarchy exploitation wins on smooth low-D redundant landscapes (robot IK, Part 2); PSO's continuous drift wins on our coverage surface. *Match the operator to the landscape.*
3. **Real problems are not uniformly random.** NFL's uniform prior includes mostly incompressible, structureless functions. Real objectives have exploitable regularity (locality, gradient-of-fitness correlation, decomposability). Metaheuristics work *in practice* precisely because they encode weak, broadly-correct priors (nearby solutions have correlated fitness — §1.7.2).
4. **The engineering job is choosing a bias that fits the problem; no algorithm wins everywhere.** This is what justifies the "choose the algorithm to fit the problem" methodology of Part 4.

Caveat (sharpened NFL). Later work (e.g. Igel & Toussaint 2003; Auger & Teytaud 2010) shows NFL holds exactly only for sets of functions **closed under permutation** of the domain; over structured subsets some algorithms *do* dominate. This does not rescue a "best algorithm," but it explains why structure-exploiting heuristics beat blind search on real data.

1.4 1.4 Exploration vs. exploitation (diversification vs. intensification)

1.4.1 1.4.1 Definitions

- **Exploration / diversification**: sampling far from current incumbents to discover new basins of attraction; increases population **diversity** and the chance of escaping local optima; slows refinement.
- **Exploitation / intensification**: concentrating samples around the best-known solutions to refine them; accelerates convergence; risks **premature convergence** to a local optimum if diversity dies first.

Formally, let $D(P_t)$ be a diversity measure, e.g. mean distance to the population centroid $\bar{x}_t$:

$$D(P_t) = \frac{1}{\mu}\sum_{i=1}^{\mu}\left\|x_t^{(i)} - \bar{x}_t\right\|_2.$$

Healthy search shows $D(P_t)$ **high early** (exploration) decaying **monotonically** to ≈ 0 (exploitation/convergence). *Our Lab-3 `diversity.png` plots exactly this collapse* — it is empirical evidence the balance was tuned correctly, not left at library defaults.

1.4.2 1.4.2 The balance is a schedule, and every algorithm has a knob

The state of the art controls the balance by **annealing** a single parameter from "explorative" to "exploitative" over the run:

Algorithm	Balance mechanism	Explorative $\to$ exploitative
PSO	inertia weight w	w linearly $0.9 \to 0.4$ (Shi & Eberhart 1998); large w = momentum/roaming, small w = local damping
GA	mutation rate p_m, selection pressure	high p_m/low pressure early; anneal p_m down
GWO	coefficient a	a linearly $2 \to 0$; $\lvert A\rvert>1$ diverge (explore), $\lvert A\rvert<1$ converge (exploit)
WOA	a (same) + spiral/encircle switch p	$a : 2 \to 0$; probabilistic switch keeps both modes
ACO	evaporation ρ	high ρ forgets, keeps exploring; low ρ locks in trails
FA	randomization α, absorption γ	anneal α; γ sets attraction range ($\gamma \to 0$ global, $\gamma \to \infty$ local)
CS	discovery prob. p_a, Lévy scale α	Lévy heavy tails give rare long explorative jumps throughout
ABC	`limit` (abandonment)	scouts re-inject exploration when a source stagnates

Our PSO uses the inertia schedule $w_t = w_{\max} - (w_{\max} - w_{\min})\, t/(T-1)$ — the direct answer to "how did you avoid premature convergence?"

1.5 1.5 Diversity maintenance & premature convergence

Premature convergence = the population collapses onto a single (sub-optimal) basin while budget remains; $D(P_t) \to 0$ *before* $x^\star$ is found, after which variation can no longer escape (crossover of near-identical parents is a no-op; PSO velocities $\to 0$). Signs (from the course slides): population becomes uniform; no best-fitness improvement over many generations. Mechanisms that counter it:

- **Mutation / immigration.** Non-zero mutation guarantees global reachability (§1.1.3). Random *immigration* injects fresh individuals when stagnation is detected — a hard reset of diversity.
- **Controlled elitism.** Keep only a *small* elite fraction; retaining too many clones the population.
- **Selection-pressure control.** Tournament size, rank-based instead of fitness-proportional selection (bounds the takeover rate so one super-individual cannot dominate in a few generations).
- **Niching / fitness sharing** (Goldberg & Richardson 1987). Penalize crowding so multiple optima coexist. Shared fitness:

$$f_i' = \frac{f_i}{\sum_{j=1}^{\mu}\mathrm{sh}(d_{ij})}, \qquad \mathrm{sh}(d) = \begin{cases} 1 - (d/\sigma_{\text{share}})^{\alpha}, & d < \sigma_{\text{share}} \\ 0, & d \geq \sigma_{\text{share}} \end{cases}$$

where $d_{ij} = \|x_i - x_j\|$ and σ_{share} is the niche radius. Individuals in a crowd share (dilute) their fitness, so selection spreads them across niches.
- **Crowding / restricted replacement** (De Jong; Deb & Goldberg 1989): offspring replace the *most similar* parent, preserving spatial spread.
- **Velocity clamping / re-init** (PSO): clamp $|v| \le v_{\max}$ to stop explosion (over-exploration) *and* GCPSO-style re-randomization of `gbest` to stop stagnation (over-exploitation).

Our optimizer relies on: velocity clamping ($v_{\max} = 0.2\times$range), the inertia schedule, and the swarm's own stochastic r_1, r_2; the `diversity.png` curve is the evidence that these keep $D(P_t) > 0$ through the exploration phase.

1.6 1.6 Mathematical definitions of convergence

1.6.1 1.6.1 What "convergence" means

Let $x_t^{\text{best}} = \arg\min_{i,\, s\le t} f(x_s^{(i)})$ be the best-so-far and $f^\star = f(x^\star)$.

- **Convergence in value (in probability):** $\forall\varepsilon > 0,\ \Pr\left(f(x_t^{\text{best}}) - f^\star > \varepsilon\right) \xrightarrow{t\to\infty} 0.$
- **Almost-sure convergence:** $\Pr\left(\lim_{t\to\infty} f(x_t^{\text{best}}) = f^\star\right) = 1.$
- **Global-search guarantee:** the (weaker, prerequisite) property that every region of positive measure is sampled infinitely often, so $x^\star$ is *reached* with probability 1. Global reachability + elitism $\Rightarrow$ a.s. convergence.

Note the distinction between converging to a **fixed point** (the swarm agrees — *stagnation*, which may be sub-optimal) and converging to the **global optimum** (the object we actually want). Conflating them is a classic viva trap (§Part 3).

1.6.2 1.6.2 Elitist GA converges; canonical GA does not

Source. G. Rudolph, "Convergence Analysis of Canonical Genetic Algorithms," *IEEE Transactions on Neural Networks* 5(1):96–101, 1994, DOI 10.1109/72.265964.

Modeling the GA as a homogeneous Markov chain on $\{0,1\}^{\ell\mu}$, Rudolph proves: the **canonical GA** (fitness-proportional selection + crossover + bit mutation, *no elitism*) does **not** converge to the global optimum — its best-so-far can oscillate because selection can lose the optimum once found. Adding **elitism** (maintain the best individual across generations) yields a chain whose best-so-far is monotone and **converges to the global optimum with probability 1**. Takeaway: *the convergence guarantee comes from elitism + positive mutation, not from crossover.*

1.6.3 1.6.3 PSO can stagnate at a non-optimum

Source. F. van den Bergh, "An Analysis of Particle Swarm Optimizers," PhD thesis, Univ. of Pretoria, 2001; and F. van den Bergh & A. P. Engelbrecht, "A study of particle swarm optimization particle trajectories," *Information Sciences* 176(8):937–971, 2006, DOI 10.1016/j.ins.2005.02.003.

Standard `gbest` PSO is **not** a global (nor even local) optimizer: once all particles coincide with `gbest`, the cognitive and social terms vanish and velocity decays as $v \to 0$, so the swarm halts at `gbest` **regardless of whether it is a local minimum** — pure *stagnation*. The **Guaranteed Convergence PSO (GCPSO)** fix forces the global-best particle to sample a shrinking random neighbourhood, restoring convergence to a local minimizer. Clerc & Kennedy (2002, DOI 10.1109/4235.985692) give the *constriction* analysis that yields stable trajectories via $\chi \approx 0.7298,\ c_1{=}c_2{=}2.05$; the inertia form $w{\in}[0.4, 0.9],\ c_1{=}c_2{\approx}1.49$ we use is its practical equivalent.

The honest position for the viva: PSO offers **no** guarantee of global optimality; we mitigate stagnation empirically (diversity monitoring, multiple seeds, matched-budget baselines, Wilcoxon test) rather than claim a proof we do not have.

1.7 1.7 Topology of the fitness landscape

A landscape is the triple $(\mathcal{S}, \mathcal{N}, f)$ where $\mathcal{N}$ is a neighbourhood/move structure. Its geometry dictates which operator wins (NFL, §1.3).

1.7.1 1.7.1 Modality

The number of local optima. **Unimodal**: one optimum (gradient/hill-climbing suffices). **Multimodal**: many (population search + niching needed). Formally x is a local minimum if $f(x) \leq f(y)\ \forall y \in \mathcal{N}(x)$. Our coverage surface is multimodal: distinct AP configurations (which AP covers which wing) are separate basins of comparable quality.

1.7.2 1.7.2 Ruggedness (fitness correlation)

How fast fitness de-correlates along a walk. Take a random walk $x_0, x_1, \ldots$ over $\mathcal{N}$ and the series $f_t = f(x_t)$; its **autocorrelation** at lag s is

$$\rho(s) = \frac{\mathbb{E}\big[(f_t - \bar{f})(f_{t+s} - \bar{f})\big]}{\mathrm{Var}(f)}, \qquad \ell = -\frac{1}{\ln|\rho(1)|}$$

(Weinberger 1990, *Biological Cybernetics* 63:325–336). Large **correlation length** $\ell \Rightarrow$ smooth, "nearby solutions have similar fitness" $\Rightarrow$ local search is informative. Small $\ell \Rightarrow$ rugged, needle-like $\Rightarrow$ near-random, population diversity is essential. Ruggedness is *why* metaheuristics encode the locality prior NFL says you need.

1.7.3 1.7.3 Neutrality

Extended regions of **equal** fitness — *neutral networks* — over which selection is blind and only drift moves the population (Kimura's neutral theory, ported to EC by Barnett, Reidys & Stadler). A landscape with plateaus starves gradient and fitness-proportional selection of signal; the search must *drift* (mutation) across the plateau to its edge. Our integer wall-crossing term creates exactly such plateaus (moving an AP within a shadow-free region does not change the crossing count), so f is locally flat in patches — another reason gradient methods stall and population drift helps.

1.7.4 1.7.4 Deceptiveness

A landscape is **deceptive** (Goldberg 1989, *Genetic Algorithms in Search, Optimization, and Machine Learning*) when low-order statistics point *away* from the global optimum: the average fitness of the schemata (building blocks) containing $x^\star$ is *lower* than that of competing schemata, so selection actively drives the population away from $x^\star$. Deceptive problems break gradient methods *and* naive GAs; they motivate linkage learning and diversity preservation. Most real landscapes are partially deceptive in places.

1.7.5 1.7.5 Why population metaheuristics fit non-convex / non-diff / multimodal / high-D / black-box

- **Non-differentiable**: they use only f values, never ∇f — immune to kinks, `max`, thresholds, integer counts (our objective has all four).
- **Non-convex / multimodal**: a *population* samples many basins in parallel; selection + variation perform implicit basin-hopping; niching keeps optima co-resident.
- **High-dimensional**: no $\binom{G}{k}$ enumeration; the swarm follows fitness-correlation structure (§1.7.2) rather than gridding, so cost scales with iterations, not with $\dim(\mathcal{S})$ exponentially. (This is precisely why our PSO beats Grid Search on the 6-D placement problem under a matched budget.)
- **Black-box**: they need only the oracle; no model, gradient, or convexity.
- **Robust to noise/neutrality**: population averaging and drift tolerate flat/noisy regions that stall single-point deterministic methods.

1.8 1.8 The eight algorithms — mechanics, one by one

Common notation: x = a solution/position; f = objective; t = iteration; $r, r_1, r_2 \sim U(0,1)$ fresh randoms; boldface = vectors.

1.8.1 1.8.1 Genetic Algorithm (GA)

- **Analogy.** Darwinian evolution: a population of chromosomes; fitter individuals reproduce more; crossover recombines parental "genes," mutation introduces novelty; over generations, good building blocks (schemata) proliferate.
- **Core equations.** Fitness-proportional (roulette) selection probability

$$p_i = \frac{f_i}{\sum_{j=1}^{\mu} f_j}.$$

Single-point crossover exchanges the tail substrings of two parents past a random cut; bit-flip mutation flips each gene independently with probability p_m. The **Schema Theorem** bounds building-block growth:

$$\mathbb{E}[m(H,t{+}1)] \;\ge\; m(H,t)\,\frac{f(H)}{\bar{f}}\Big[1 - p_c\,\frac{\delta(H)}{\ell-1} - o(H)\,p_m\Big],$$

where $m(H,t)$ = instances of schema H at gen t, $f(H)$ = mean fitness of H, $\bar{f}$ = population mean fitness, $\delta(H)$ = defining length, $o(H)$ = order, ℓ = string length, p_c = crossover rate. Short, low-order, above-average schemata grow exponentially.
- **Parameters & sensitivity.** Population $\mu \in [30, 100]$; crossover $p_c \in [0.6, 0.95]$; mutation $p_m \in [0.001, 0.01]$ (values > 0.05 usually harmful — devolves to random walk). Selection scheme (tournament/rank) controls takeover speed; encoding choice dominates everything.
- **Best-fit landscape.** Combinatorial / discrete / mixed spaces with meaningful recombination structure (feature selection, scheduling, permutations).
- **Signature strength.** Extreme representational flexibility — encode almost anything (binary, permutation, tree, real vector) and design matched operators.
- **Main limitation.** Premature convergence via diversity loss; performance highly sensitive to encoding and operator design.
- **Seminal cite.** J. H. Holland, *Adaptation in Natural and Artificial Systems*, Univ. of Michigan Press, 1975 (2nd ed. MIT Press 1992, DOI 10.7551/mitpress/1090.001.0001).

1.8.2 1.8.2 Particle Swarm Optimization (PSO)

- **Analogy.** A flock/school foraging: each bird (particle) balances its own memory of the best food it found (`pbest`, nostalgia) against the flock's best (`gbest`, social influence), with momentum (inertia) carrying it forward.
- **Core equations** (inertia form; the course slide's rule):

$$\mathbf{v}_i^{t+1} = w\,\mathbf{v}_i^t + c_1 r_1(\mathbf{p}_i - \mathbf{x}_i^t) + c_2 r_2(\mathbf{g} - \mathbf{x}_i^t), \qquad \mathbf{x}_i^{t+1} = \mathbf{x}_i^t + \mathbf{v}_i^{t+1}.$$

$\mathbf{v}_i$ = velocity, $\mathbf{x}_i$ = position, $\mathbf{p}_i$ = personal best, $\mathbf{g}$ = global best, w = inertia, c_1 = cognitive (personal) coefficient, c_2 = social coefficient, $r_1, r_2 \sim U(0,1)$. Inertia w was added by Shi & Eberhart 1998 (DOI 10.1109/ICEC.1998.699146); it is absent from the 1995 original.
- **Parameters & sensitivity.** $w \in [0.4, 0.9]$ (often linearly decreasing); c_1,$c_2 \approx 2.0$ (range 0.5–2.5); swarm 20–50; velocity clamp $v_{\max}$. Balance is very sensitive to w and to the c_1:c_2 ratio; w too large $\Rightarrow$ divergence, too small $\Rightarrow$ premature convergence. Clerc–Kennedy constriction (χ=0.7298) is the stability-guaranteed alternative.
- **Best-fit landscape.** Continuous, real-valued, moderately multimodal optimization — *our Wi-Fi placement is a textbook fit.*
- **Signature strength.** Few operators, few parameters, fast empirical convergence on continuous problems; trivially parallel.
- **Main limitation.** Stagnation/velocity collapse at a non-guaranteed point (van den Bergh 2001); weaker on combinatorial spaces without re-encoding.
- **Seminal cite.** J. Kennedy & R. C. Eberhart, "Particle Swarm Optimization," *Proc. IEEE ICNN'95*, vol. 4, pp. 1942–1948, DOI 10.1109/ICNN.1995.488968.

1.8.3 1.8.3 Ant Colony Optimization (ACO)

- **Analogy.** Ants deposit pheromone on trails; shorter routes are traversed faster and reinforced more; evaporation erases stale trails; the colony collectively converges on short paths — *stigmergy* (indirect coordination via the environment).
- **Core equations.** Probabilistic edge choice by ant k at node i:

$$p_{ij}^k = \frac{\tau_{ij}^\alpha \, \eta_{ij}^\beta}{\sum_{l \in \mathcal{N}_i^k} \tau_{il}^\alpha \, \eta_{il}^\beta}, \qquad \tau_{ij} \leftarrow (1-\rho)\, \tau_{ij} + \sum_k \Delta\tau_{ij}^k, \quad \Delta\tau_{ij}^k = \tfrac{Q}{L_k}.$$

  τ_{ij} = pheromone on edge (i,j), η_{ij} = heuristic desirability ($\approx 1/d_{ij}$), α, β = weight exponents, $\mathcal{N}_i^k$ = feasible next nodes, ρ = evaporation rate, Q = constant, L_k = tour length of ant k.
- **Parameters & sensitivity.** $\alpha \approx 1$, $\beta \in [2,5]$, $\rho \approx 0.5$, ants $m \approx n$, Q. Over-emphasizing τ (large α, small ρ) causes early lock-in; MAX–MIN Ant System (Stützle & Hoos 2000) bounds $\tau \in [\tau_{\text{min}}, \tau_{\text{max}}]$ to prevent it.
- **Best-fit landscape.** Discrete graph/combinatorial construction problems: routing, TSP, scheduling, sequencing.
- **Signature strength.** Constructs solutions incrementally with an explicit *learned* memory (pheromone) — excellent on shortest-path/graph problems and dynamic re-routing.
- **Main limitation.** Pheromone stagnation / premature convergence; sensitive to ρ, α, β; mostly discrete.
- **Seminal cite.** M. Dorigo, V. Maniezzo, A. Colorni, "Ant System: Optimization by a Colony of Cooperating Agents," *IEEE Trans. SMC-B* 26(1):29–41, 1996, DOI 10.1109/3477.484436 (concept: Dorigo PhD thesis, Politecnico di Milano, 1992).

1.8.4 1.8.4 Artificial Bee Colony (ABC)

- **Analogy.** Honeybee foraging with a division of labour: *employed* bees exploit known food sources, *onlookers* probabilistically pick richer sources to exploit further, *scouts* abandon exhausted sources and explore randomly.
- **Core equations.** Employed/onlooker neighbour step and onlooker selection:

$$v_{ij} = x_{ij} + \phi_{ij}\,(x_{ij} - x_{kj}), \quad \phi_{ij} \in [-1,1], \qquad p_i = \frac{\text{fit}_i}{\sum_{n=1}^{SN} \text{fit}_n}.$$

  x_{ij} = dimension j of source i; x_{kj} = a random other source $k \neq i$; ϕ_{ij} = uniform random; SN = number of food sources. A source not improved within `limit` trials is abandoned and re-initialized (scout): $x_{ij} = x_j^{\text{min}} + r\,(x_j^{\text{max}} - x_j^{\text{min}})$.
- **Parameters & sensitivity.** Colony/source count SN; abandonment `limit` (governs explore/exploit — small `limit` over-explores); employed=onlookers=$SN/2$.
- **Best-fit landscape.** Continuous, multimodal numerical optimization.
- **Signature strength.** Strong global exploration via scouts; only one real control parameter (`limit`) beyond size.
- **Main limitation.** Good exploration but *poor exploitation* — slow final convergence near optima (motivating GABC and hybrids).
- **Seminal cite.** D. Karaboga, "An Idea Based on Honey Bee Swarm for Numerical Optimization," Tech. Rep. TR06, Erciyes Univ., 2005; Karaboga & Basturk, *J. Global Optimization* 39(3):459–471, 2007, DOI 10.1007/s10898-007-9149-x.

1.8.5 1.8.5 Firefly Algorithm (FA)

- **Analogy.** Fireflies attract mates by bioluminescence; a dimmer firefly moves toward a brighter one; perceived brightness falls with distance (light absorption), so attraction is local — producing automatic subswarming around multiple optima.
- **Core equations.** Attractiveness and movement:

$$\beta(r) = \beta_0\, e^{-\gamma r^2}, \qquad \mathbf{x}_i \leftarrow \mathbf{x}_i + \beta_0\, e^{-\gamma r_{ij}^2}(\mathbf{x}_j - \mathbf{x}_i) + \alpha\, \varepsilon_i.$$

  β_0 = attractiveness at $r{=}0$; γ = light-absorption coefficient; $r_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|$; α = randomization scale; ε_i = random vector.

- **Parameters & sensitivity.** $\alpha \approx 0.2$ (often annealed); $\beta_0 = 1$; $\gamma \in [0.1, 10]$: $\gamma \to 0$ makes it PSO-like (global), $\gamma \to \infty$ makes it a random walk. γ is the critical knob.
- **Best-fit landscape.** Highly **multimodal** continuous problems where finding *many* optima matters (the local-attraction property partitions the swarm into niches).
- **Signature strength.** Automatic multimodal subdivision — locates multiple optima simultaneously without explicit niching.
- **Main limitation.** Premature convergence/stagnation on high-D problems; $O(\mu^2)$ pairwise comparisons per iteration.
- **Seminal cite.** X.-S. Yang, "Firefly Algorithms for Multimodal Optimization," *SAGA 2009*, LNCS 5792:169–178, DOI 10.1007/978-3-642-04944-6_14 (origin: *Nature-Inspired Metaheuristic Algorithms*, Luniver Press, 2008).

1.8.6 1.8.6 Grey Wolf Optimizer (GWO)

- **Analogy.** Grey-wolf pack hierarchy and hunting: the three best solutions are the α, β, δ wolves; the rest (ω) update their positions by encircling the estimated prey (optimum) as directed by the three leaders.
- **Core equations.**

$$\mathbf{D} = |\mathbf{C} \cdot \mathbf{X}_p - \mathbf{X}|, \qquad \mathbf{X}(t+1) = \mathbf{X}_p - \mathbf{A} \cdot \mathbf{D},$$

  with leader aggregation $\mathbf{X}_1 = \mathbf{X}_\alpha - \mathbf{A}_1 \mathbf{D}_\alpha$, $\mathbf{X}_2 = \mathbf{X}_\beta - \mathbf{A}_2 \mathbf{D}_\beta$, $\mathbf{X}_3 = \mathbf{X}_\delta - \mathbf{A}_3 \mathbf{D}_\delta$, and $\mathbf{X}(t+1) = \frac{1}{3}(\mathbf{X}_1 + \mathbf{X}_2 + \mathbf{X}_3)$. Coefficients $\mathbf{A} = 2\mathbf{a} \cdot \mathbf{r}_1 - \mathbf{a}$, $\mathbf{C} = 2\mathbf{r}_2$, where **a** decreases **linearly from 2 to 0** across iterations; $|\mathbf{A}|>1$ = explore, $|\mathbf{A}|<1$ = exploit.
- **Parameters & sensitivity.** Population (≈30), max-iterations, and the **a** schedule (the sole explore/exploit control). Few parameters ⇒ easy to tune, but the linear **a** schedule is a rigid prior.
- **Best-fit landscape.** Smooth-ish, low-to-moderate-dimensional, *redundant* continuous problems (e.g. robot inverse kinematics — Part 2) where strong late-stage exploitation pays.
- **Signature strength.** Excellent exploitation with almost no parameters; the three-leader consensus damps premature lock to a single incumbent.
- **Main limitation.** Premature convergence on high-D multimodal problems (no mutation/diversity operator; driven by the leaders).
- **Seminal cite.** S. Mirjalili, S. M. Mirjalili, A. Lewis, "Grey Wolf Optimizer," *Advances in Engineering Software* 69:46–61, 2014, DOI 10.1016/j.advengsoft.2013.12.007.

1.8.7 1.8.7 Cuckoo Search (CS)

- **Analogy.** Brood parasitism: cuckoos lay eggs in host nests; the best eggs (solutions) survive; hosts discover and discard a fraction p_a of alien eggs (abandonment); new nests are sought by **Lévy flights** — heavy-tailed random walks that mix frequent short hops with rare long jumps.
- **Core equations.**

$$\mathbf{x}_i^{t+1} = \mathbf{x}_i^t + \alpha \oplus \text{Lévy}(\lambda), \qquad \text{Lévy}(\lambda) \sim u = s^{-\lambda},\ 1 < \lambda \leq 3,$$

  where α = step-size scaling, $\oplus$ = entrywise product, and a fraction p_a of worst nests is replaced by a biased random walk $\mathbf{x}_i^{t+1} = \mathbf{x}_i^t + r(\mathbf{x}_j^t - \mathbf{x}_k^t)$.
- **Parameters & sensitivity.** Discovery probability $p_a \approx 0.25$ (authors report low sensitivity), Lévy exponent $\lambda \approx 1.5$, nests $n \in [15, 40]$. Notably *few* parameters and robust defaults.
- **Best-fit landscape.** Multimodal continuous global optimization; the Lévy heavy tail gives persistent long-range exploration that resists premature convergence.
- **Signature strength.** Lévy-flight global exploration escapes local optima that Gaussian-step methods miss; very few parameters.
- **Main limitation.** Fixed Lévy step size ⇒ slow late convergence on complex/high-D problems.
- **Seminal cite.** X.-S. Yang & S. Deb, "Cuckoo Search via Lévy Flights," *World Congress on Nature & Biologically Inspired Computing (NaBIC 2009)*, IEEE, pp. 210–214, DOI 10.1109/NABIC.2009.5393690.

1.8.8 1.8.8 Whale Optimization Algorithm (WOA)

- **Analogy.** Humpback-whale *bubble-net* feeding: whales encircle prey and swim along a shrinking spiral while blowing bubbles; the algorithm alternates between shrinking-encircle and logarithmic-spiral moves toward the best solution.
- **Core equations.** Encircle vs. spiral, chosen by a coin flip p:

$$\mathbf{X}(t+1) = \begin{cases} \mathbf{X}^* - \mathbf{A} \cdot |\mathbf{C}\,\mathbf{X}^* - \mathbf{X}|, & p < 0.5 \\ \mathbf{D}'\, e^{bl} \cos(2\pi l) + \mathbf{X}^*, & p \geq 0.5 \end{cases}$$

  where $\mathbf{X}^*$ = best solution so far, $\mathbf{D}' = |\mathbf{X}^* - \mathbf{X}|$, b = spiral shape constant, $l \in [-1, 1]$, $\mathbf{A} = 2\mathbf{ar} - \mathbf{a}$, $\mathbf{C} = 2\mathbf{r}$, and $\mathbf{a}$ decreases linearly $2 \to 0$. When $|\mathbf{A}| \geq 1$ the encircle term targets a *random* whale instead of $\mathbf{X}^*$ (exploration).
- **Parameters & sensitivity.** Population, max-iterations, spiral constant b, and the $\mathbf{a}$ schedule; like GWO, very few knobs.
- **Best-fit landscape.** Continuous global optimization; the spiral gives a distinctive exploitation trajectory useful on engineering design problems.
- **Signature strength.** The dual encircle/spiral operator balances explore/exploit with minimal parameters; strong on many engineering benchmarks.
- **Main limitation.** Slow convergence and local-optima trapping on complex high-D problems; diversity loss in late iterations.
- **Seminal cite.** S. Mirjalili & A. Lewis, "The Whale Optimization Algorithm," *Advances in Engineering Software* 95:51–67, 2016, DOI 10.1016/j.advengsoft.2016.01.008.

1.9 1.9 Side-by-side summary table

Algorithm	Inspiration	Core update (essence)	Key parameters	Best-fit landscape	Signature strength	Main limitation	Seminal cite
GA	Darwinian evolution	roulette $p_i{=}f_i/\sum f$; crossover; bit-flip mutation p_m	μ 30–100, p_c 0.6–0.95, p_m 0.001–0.01	discrete/combinatorial/mixed w/ recombination structure	representation flexibility (encode anything)	premature convergence; encoding-sensitive	Holland 1975
PSO	bird flock / fish school	$\mathbf{v}{=}w\mathbf{v}{+}c_1r_1(\mathbf{p}{-}\mathbf{x}){+}c_2r_2(\mathbf{g}{-}\mathbf{x})$	w 0.4–0.9, c_1,$c_2{\approx}2$, swarm 20–50	continuous, moderately multimodal	few params, fast on continuous, parallel	stagnation, no global guarantee	Kennedy & Eberhart 1995
ACO	ant stigmergy (pheromone)	$p{\propto}\tau^\alpha\eta^\beta$; $\tau{\leftarrow}(1{-}\rho)\tau{+}\sum Q/L_k$	$\alpha{\approx}1$, β 2–5, $\rho{\approx}0.5$	discrete graph/routing/sequencing	incremental construction w/ learned memory	pheromone stagnation; mostly discrete	Dorigo et al. 1996
ABC	honeybee foraging roles	$v{=}x{+}\phi(x{-}x_k)$; scouts after `limit`	SN, `limit`	continuous multimodal	strong exploration; one main knob	weak exploitation (slow finish)	Karaboga 2005
FA	firefly bioluminescence	$\mathbf{x}{+}{=}\beta_0e^{-\gamma r^2}(\mathbf{x}_j{-}\mathbf{x}_i){+}\alpha\varepsilon$	$\alpha{\approx}0.2$, $\beta_0{=}1$, γ 0.1–10	highly multimodal (many optima)	automatic multimodal niching	premature convergence; $O(\mu^2)$/iter	Yang 2008/2009
GWO	grey-wolf pack hierarchy	3 leaders $\alpha\beta\delta$; $\mathbf{X}{=}\mathbf{X}_p{-}\mathbf{AD}$; a:2$\to$0	pop, iters, a schedule	smooth/redundant low–mid-D continuous	strong exploitation, few params	high-D multimodal premature convergence	Mirjalili et al. 2014

Algorithm	Inspiration	Core update (essence)	Key parameters	Best-fit landscape	Signature strength	Main limitation	Seminal cite
CS	cuckoo brood parasitism	$\mathbf{x}{+}{=}\alpha \oplus$ Lévy(λ); abandon p_a	$p_a{\approx}0.25$, $\lambda{\approx}1.5$, n 15–40	multimodal continuous global	Lévy long-jumps escape local optima	fixed step ⇒ slow finish	Yang & Deb 2009
WOA	humpback bubble-net	encircle/spiral switch; a:2→0	pop, iters, b, a schedule	continuous engineering-design global	dual operator, minimal params	slow, local-optima trap on high-D	Mirjalili & Lewis 2016

Cross-cutting reading. All eight are the §1.1.2 template with different $(\mathcal{V}, \mathcal{Q}, M)$: GA/ABC use explicit selection + recombination/neighbour variation; PSO/FA/GWO/WOA use *attraction toward memory* (pbest/gbest/brighter/leaders/best) as implicit selection; ACO uses environmental memory (pheromone). All expose exactly one dominant explore/exploit knob (§1.4.2). None escapes NFL (§1.3): each is a *bet* that the target landscape matches its operator's locality/attraction prior — the bet we place, and justify, for our Wi-Fi problem in Part 4.